Technical Note

Power Electronics and Control Systems

Three-level Vienna rectifier on the 690 V grid

`vienna_rectifier`: power stage and control, fixes of November 2026, design check of the loops, start-up, steady state and open points

The model `vienna_rectifier` of `pwr-control/modelization-and-control` after the revision of November 2026: what it contains, why it did not run, what was changed in the model and in its initialization, how its loops are designed, and how it behaves at the start and in steady state, checked with linear models and with a switched replica of the model.

Authors:	**Revision:**	**Date:**
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Abstract: The model `vienna_rectifier` feeds a 250 kW three-level Vienna rectifier from a 1 MV A, 690 V grid emulator through an LCL filter whose grid side is the leakage of the Dyn11 transformer; the split DC link (2×14.4 mF) supplies a 5 Ω load, 229 kW at 1070 V. The control is the dq control of the AFE models of the repository: DDSRF-PLL, DC-voltage PI, vector PI of the converter current, space vector modulation and three-level PWM at 4 kHz with double update. As saved, the model did not run: no initialization gave the `param.*` names that the grid emulator read, the current PI read the gains of a PMSM inverter, a bus object and the names of the modulator lines were missing, 23 From blocks had no Goto, and the DDSRF-PLL was fed with the negative sequence and could not lock (1383 A peaks and 1916 V on the DC link in the switched replica). This note describes the model, the new initialization with the struct `param` and the changes to the model, and checks the design: operating point (8.7 % of voltage margin), LCL resonance at 1381 Hz, just above $f_s/6$, current loop (135 Hz, 70° of phase margin), DC-voltage loop (21 Hz, 98°), DDSRF-PLL with its decoupling filter in the loop (58°), neutral-point ripple (5.4 V peak to peak). A switched replica of the model gives the start-up (752 A peak, DC link between 758 and 1171 V) and the steady state (THD of the grid current 3.4 %). The start-up peak comes from the anti-windup of the DC-voltage PI, which drives the current reference positive while the DC voltage is still above its reference, and from the current loop, which has no feed-forward of the grid voltage. Changes quantified but not applied: a DC-voltage PI with lower limit 0 and conditional integration (456 A, no overshoot), or a feed-forward of the capacitor voltage with a DC-voltage reference that starts from the measured voltage (455 A, no overshoot); a zero sequence that follows the sign of the current (THD 2.5 %).



Revision history

Rev.	Date	Description	Authors
00	November 2026	First issue: model, changes of November 2026, design check, switched replica of the start-up and of the steady state, open points; scripts `py/vienna_sim.py` and `py/vienna_design_check.py`.	Davide Bagnara

Project Repository: modelization-and-control.git (vienna_rectifier)

Framework: MATLAB/Simulink, Python



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Nomenclature

Symbol	Meaning
AFE	active front end: the grid-side converter with its control
leg, pole	one phase of the bridge; its AC terminal (the pole)
np	neutral point: the middle point of the DC link, between C_1 and C_2
D_1, D_4, Q_2, Q_3	diodes and switches of a Vienna leg, with the names of the library block
u_{dc}, v_{C1}, v_{C2}	DC-link voltage and the voltages of its upper and lower half
C_1, C_2, C_{ser}, C_{np}	capacitors of the two halves (14.4 mF each); $C_{\mathrm{ser}} = C_1C_2/(C_1 + C_2)$ seen by u_{dc}, $C_{\mathrm{np}} = C_1 + C_2$ seen by the neutral point
m, m_{eff}	pole reference of a leg, $-1 \le m \le 1$ (average pole voltage $m\,u_{\mathrm{dc}}/2$), and the reference that the leg can follow
u	output of the current PI: the converter voltage in pu of $u_{\mathrm{dc}}/\sqrt{3}$ (phase peak); $\lvert u\rvert \le 1$ is the linear range of the modulation
i_1, i_2, v_{c}	converter current, grid current (at the secondary of the transformer), capacitor voltage of the filter
L_1, R_1, C_{f}, R_{c}	converter-side inductor and its resistance, filter capacitor (in star) and its series resistor
L_2, R_2	grid side of the LCL filter: leakage inductance and copper resistance of the transformer, referred to the secondary
E	no-load phase voltage (peak) of the grid emulator at the secondary of the transformer
u_{bez}, i_{bez}	per-unit bases of the control: 563.4 V (phase peak of 690 V) and 328.7 A (peak)
ω, f_0	grid pulsation $2\pi f_0$, $f_0 = 50\,\mathrm{Hz}$
t_{s}, f_{s}	control period 125 µs (PWM at 4 kHz, double update) and control frequency $1/t_{\mathrm{s}} = 8\,\mathrm{kHz}$
t_{c}	step of the simulation, $t_{\mathrm{s}}/200 = 0.625\,\mu\mathrm{s}$
$\xi\eta$	rotating frame of the model (the dq frame), aligned with the angle of the PLL
k_{p}, k_{i}	proportional and integral gain of a PI; the loop is named in the text
δ	angle by which the converter voltage lags the converter current
ρ	$e^{j\omega t_{\mathrm{s}}}$: the pole of the dq integrator seen in the stationary frame
τ	time constant of the low-pass filters of the DDSRF-PLL, $1/\omega$
i_{np}	current into the neutral point
z	variable of the z transform, $z = e^{j2\pi f t_{\mathrm{s}}}$ on the unit circle
PM, GM	phase margin and gain margin of a loop
THD	total harmonic distortion, harmonics up to the 40th, in % of the fundamental

1 Introduction

The model `vienna_rectifier.slx` in `foundations_of_power_electronics/vienna_rectifier` of [1] is a 250 kW three-level Vienna rectifier [4, 5] on a 690 V grid. It uses the grid emulator, the LCL filter, the control and the three-level modulator of the AFE models of the repository, with the Vienna bridge of the library [2] in place of the T-type bridge. As saved, the model did not run, and its initialization was the shared template of the AFE and inverter models. In November 2026 the initialization was rewritten as one struct `param` with the functions of `private/`, the model was re-pointed to `param.*` and its broken parts were fixed.

This note:

- describes the model: the power stage, the Vienna leg and what it can and cannot do, the modulator and the control (Section 2);
- lists what did not work and what was changed, in the model and in its initialization (Section 3);
- checks the design with linear models: operating point, LCL resonance, current loop, DC-voltage loop, DDSRF-PLL, Vienna constraint and neutral point (Section 4);
- shows the start-up and the steady state of a switched replica of the model, with a few options (Section 5);
- lists the open points and the recommended changes (Section 6) and the files (Section 7).

How the results were obtained. No MATLAB was available for this revision. The changes to the model were made in the XML of the `.slx` file and checked with Octave and stubs of the Simulink functions: the initialization runs, all 93 `param.*` references of the active blocks resolve and every active From block reaches its Goto. The model was then opened in MATLAB, where a scoped Goto of the DDSRF-PLL without its visibility block came out and was fixed (Table 3); the check of the From blocks now applies the visibility rules. The linear checks of Section 4 are computed by `private/vienna_design.m`, which the initialization runs and prints, and again by `py/vienna_design_check.py` of this note. The time responses of Section 5 come from `py/vienna_sim.py`, a switched replica of the model written step by step as the blocks and the C code of the library compute it (Section 5.1). The replica has not yet been compared with a run of the revised model: that run is the check of the numbers of Section 5.

2 The model

2.1 Power stage

Figure 1 shows the model and Table 1 its data. The grid emulator (library block `three_phase_voltage_source_with_frt_emulator`) drives three controlled voltage sources whose amplitude rises from zero with a 20 ms ramp and a 50 Hz second order filter; both fault generators of the emulator are off. The coupling transformer `3ph_Dyg` is made of three single-phase transformers: delta primary with 24 turns, grounded star secondary with 14 turns, so the no-load secondary voltage is 697 V ($E = 569.2$ V phase peak). The LCL filter has the inductors $L_{\mathrm{Fu}} = 500\,\mu\mathrm{H}$ (plus 1 µH of the measurement blocks, and 500 µH of common-mode inductance) on the converter side,

the capacitors $C_{\mathrm{Fu}} = 200\,\mu\mathrm{F}$ in star with $50\,\mathrm{m}\Omega$ in series, and on the grid side only the leakage of the transformer, $L_2 = 76.6\,\mu\mathrm{H}$ at the secondary: the equivalent grid impedance of the emulator object is not read by any block. The Vienna bridge (library `vienna_rectifier_ideal_switch_based_model`, ideal switches and diodes with conduction drops and RC snubbers) charges the split DC link `dclink_for_three_levels`; the neutral point is grounded through $100\,\mathrm{k}\Omega$ and a $5\,\Omega$ resistor across the DC link is the load. The DC link starts precharged at 1070 V, so the simulation starts with the load connected and the grid voltage at zero.

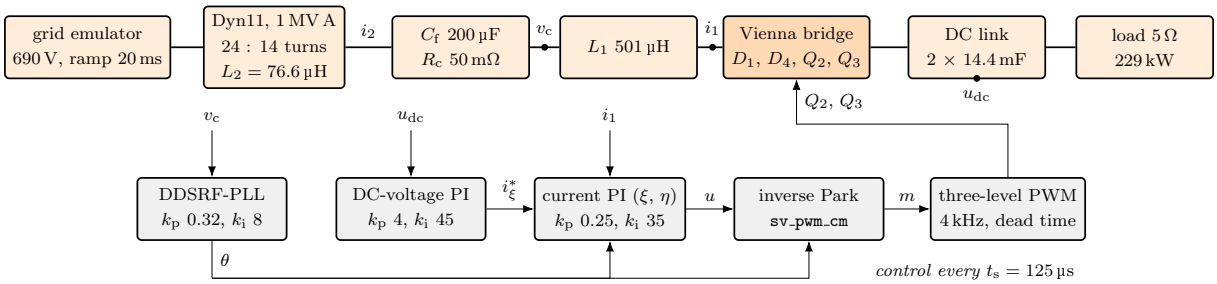


Figure 1: Structure of the model: power stage (top) and control `afe_dq_mod1/AFE_control` (bottom). The dots mark the measurements, which the control reads quantized to 12 bit and one step late. The converter-current loop is closed on i_1 in the frame of the PLL; the DC-voltage loop gives its active reference.

Table 1: Data of the model (`init_model.m` and the classes of the library it calls).

part	data
grid emulator	690 V, 50 Hz; amplitude from 0 with a 20 ms ramp and a 50 Hz second order filter; fault generators off
transformer	1 MV A, Dyn11, three single-phase units, 24 : 14 turns; u_cc 5 %, efficiency 95 %; at the secondary $L_2 = 76.6\,\mu\mathrm{H}$, $R_2 = 24.1\,\mathrm{m}\Omega$, $E = 569.2\,\mathrm{V}$ peak
LCL filter	$L_1 = 501\,\mu\mathrm{H}$, $R_1 = 3.9\,\mathrm{m}\Omega$; $C_\mathrm{f} = 200\,\mu\mathrm{F}$ in star (star point floating), $R_\mathrm{c} = 50\,\mathrm{m}\Omega$; common-mode inductance 500 µH
bridge	Vienna, ideal switches and diodes (data set `silicon_high_power_ideal_switch`): switch 1.5 mΩ, diode 2 V and 0.15 mΩ, RC snubbers 10 nF and 2.2 kΩ; dead time 3 µs
DC link	2 × 14.4 mF, measuring dividers 107 kΩ per half, neutral point to ground 100 kΩ; precharged at 1070 V
load	5 Ω across the DC link: 229 kW at 1070 V
control	$t_\mathrm{s} = 125\,\mu\mathrm{s}$; bases $u_\mathrm{bez} = 563.4\,\mathrm{V}$, $i_\mathrm{bez} = 328.7\,\mathrm{A}$, u_dc base 1070 V; ADC quantization $1/2^{11}$ pu
simulation	1.25 s; ode23tb with MaxStep $t_\mathrm{c} = 0.625\,\mu\mathrm{s}$; local solver of the grid emulator at t_c

2.2 The Vienna leg

Each leg of the bridge (Figure 2) has a diode D_1 from the pole to the positive rail, a diode D_4 from the negative rail to the pole and a bidirectional switch between the pole and the neutral point, made of two switches in common emitter, Q_3 with its collector at the pole and Q_2 with its collector at the neutral point. With the current i counted positive into the pole (rectifier

convention), the pole voltage with respect to the neutral point is

$$v_{\text{pole}} = \begin{cases} 0 & \text{if } i > 0 \text{ and } Q_3 \text{ is on, or } i < 0 \text{ and } Q_2 \text{ is on,} \\ +v_{\text{C1}} & \text{if } i > 0 \text{ and } Q_3 \text{ is off (through } D_1\text{),} \\ -v_{\text{C2}} & \text{if } i < 0 \text{ and } Q_2 \text{ is off (through } D_4\text{).} \end{cases} \tag{2.1}$$

The leg can put its pole only at the neutral point or at the rail that has the sign of its current. Two consequences follow, and both come back in the rest of the note:

- the rails take current only through the diodes, so the power goes only from the grid to the DC link: the bridge cannot give energy back, and a DC-voltage overshoot is discharged only by the load;
- where the pole reference and the current have opposite signs, the leg cannot follow the reference: its pole stays at the neutral point. This happens in a window of a few degrees after each zero crossing of the current (Section 4.6).

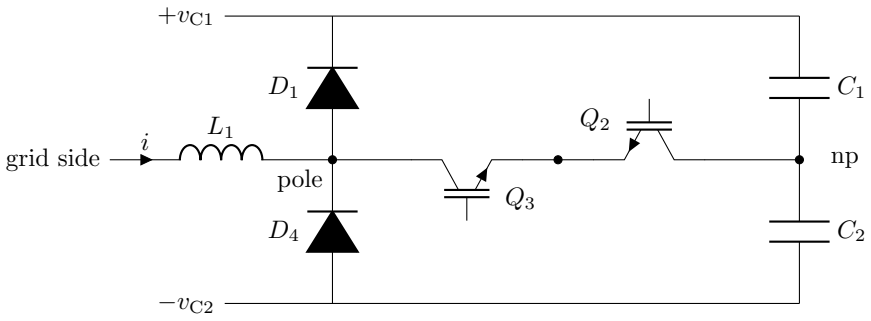


Figure 2: One leg of the Vienna bridge of the library (`vienna_rectifier_ideal_switch_based_model`) with the split DC link. Q_3 and Q_2 have anti-parallel diodes (not drawn): a positive current flows to the neutral point through Q_3 and the diode of Q_2, a negative one through Q_2 and the diode of Q_3.

2.3 Modulator

The three-level PWM of the library (`rectifier/sv_pwm`) compares the pole reference m of each leg with two carriers in phase disposition at 4 kHz, c between 0 and 1 and $c-1$ between -1 and 0, and gives the gate commands of the T-type pattern:

$$Q_1 = (m > c), \qquad Q_4 = (m < c-1), \qquad Q_3 = \overline{Q_1}, \qquad Q_2 = \overline{Q_4}, \tag{2.2}$$

each delayed at turn-on by the dead time of 3 µs. The Vienna leg uses only Q_2 and Q_3. For $m > 0$, Q_3 is off for the fraction m of the carrier period and Q_2 is always on; for $m < 0$, Q_2 is off for the fraction $|m|$ and Q_3 is always on. With (2.1), the average pole voltage over a carrier period is

$$\bar{v}_{\text{pole}} = m_{\text{eff}} \frac{u_{\text{dc}}}{2}, \qquad m_{\text{eff}} = \begin{cases} m & \text{if } m\,i \geq 0, \\ 0 & \text{if } m\,i < 0, \end{cases} \tag{2.3}$$



with equal halves of the DC link. The reference m comes from the C caller `sv_pwm_cm` (space vector modulation): from the control voltage u_x of each phase, in pu of $u_\mathrm{dc}/\sqrt{3}$,

$$m_\mathrm{x} = \frac{2}{\sqrt{3}}\,u_\mathrm{x} + v_0, \qquad v_0 = -\frac{1}{2}\Big(\max_x \frac{2}{\sqrt{3}}u_\mathrm{x} + \min_x \frac{2}{\sqrt{3}}u_\mathrm{x}\Big), \tag{2.4}$$

the min-max zero sequence, which keeps $|m| \le 1$ up to $|u| = 1$. The modulator of the model reads the output of `sv_pwm_cm` as the fraction of the high state, which inverts the sign: the pole reference of (2.2) is $-m_\mathrm{x}$ and the converter voltage is $-u$ (Section 2.4).

2.4 Control

The control (`afe_dq_mod1/AFE_control/afe_control`) is triggered by the modulator at every peak and valley of the carrier, every $t_\mathrm{s} = 125\,\mu\mathrm{s}$, and works in pu of $u_\mathrm{bez} = 563.4\,\mathrm{V}$ and $i_\mathrm{bez} = 328.7\,\mathrm{A}$:

- **Acquisitions**: capacitor voltages, converter currents and u_dc, quantized to 12 bit and one step late. The converter current goes through a gain -1: the control uses the rectifier convention. A PI aligns the local clock of the AFE (gain 1.0002) to the master clock.
- **DDSRF-PLL** on the capacitor voltage: a first harmonic tracker per phase (held in reset for the first 12.5 ms), the decoupled double synchronous reference frame with low-pass filters $\tau = 1/\omega$ [8], a PI on the normalized η component of the positive sequence ($k_\mathrm{p} = 0.32$, $k_\mathrm{i} = 8$), a frequency integrator clamped to $0.5 \ldots 1.5$ pu and started at 1 pu, the phase integrator and a double integrator observer for the frequency output. The same block, before and after its fix, is analysed in [3].
- **DC-voltage PI** (`udc_ctrl`): the reference is a step at 1 ms to 1 pu (1070 V) through a 20 ms low-pass filter with zero initial state; PI $k_\mathrm{p} = 4$, $k_\mathrm{i} = 45$ in velocity form with back-calculation; the output, limited to ± 1.65 pu, is the reference of the active current i_ξ^*.
- **Current PI** (`current_ctrl`): vector PI on the ξ and η components of the converter current, $k_\mathrm{p} = 0.25$, $k_\mathrm{i} = 35/\mathrm{s}$, reactive reference zero; the output vector is clipped to $|u| \le 1$ and the integrators hold while it is clipped. A zero-sequence PI, made for modules in parallel, is computed but multiplied by 0.
- **Modulation**: inverse Park transform with the angle of the PLL and `sv_pwm_cm` (2.4), then the three-level PWM (2.2).

There is no feed-forward of the grid voltage: the integrators of the current PI build the whole converter voltage. The two sign inversions, -1 on the measured current and $-u$ at the modulator, cancel in the loop gain: a positive error of the current gives a converter voltage that increases the current, as the switched replica confirms.



3 What did not work and what was changed

3.1 Initialization

The legacy initialization was the shared template of the AFE and inverter models (about 700 lines, most of them for parts that this model does not have) and stopped at `grid_fault_generator` with "Not enough input arguments"; the grid emulator of the model already read `param.*` names that no script defined. The new `init_model.m` is a plain-text live script that fills one struct `param` (Table 2), runs the design check of Section 4, opens the model, imports the two C types of the C callers, creates the bus object `PowerMeterIecBus` of the IEC power meter and checks the model: every `param.*` read by an active block must exist and every active From block must have a Goto. The functions of `private/` are called through the handles of `vienna_rectifier_private.m`, because the Live Editor runs a live script from a copy outside the model folder. The old script is kept as `init_model_legacy.m`.

Table 2: Groups of the struct `param` and the functions of `private/` that fill them.

group	content
`param.sim`	stop time 1.25 s, nonlinear iterations of the grid emulator, model check and design figures on or off
`param.time`, `param.glb_time`	4 kHz, double update, dead time; `timing_param` gives t_s, t_c and the other sample times
`param.grid`	1 MV A, 690 V, Dyn11, core data; `grid_param` adds the transformer data set, the magnetizing inductance and the emulator object `param.grid.emu`
`param.frt`	data of test 25.4 of VDE-AR-N 4110 (`frt_param`), not applied
`param.afe.hw`, `.dev`	250 kW hardware of the 690 V class (`afe_hw_param`), switches and diodes
`param.afe.dclink`, `param.load`	initial DC voltage, measuring dividers, neutral point to ground; load resistor
`param.afe.ctrl`, `.pll`, `.behav`	class `ctrl_afe_setup` with the gains of the model; initial value of the PLL integrator; clock gain, carrier phase, zero-sequence gain, power limit flag
`param.ctrl`, `param.meas`	clipping decay of the current PI, clock alignment PI, 50 Hz filter of the emulator; quantization and ranges of the ADC
`param.afe.design`	results of the design check `vienna_design`

3.2 Changes to the model

The model was edited in the XML of `vienna_rectifier.slx`: 158 block parameters and the three entries of the configuration set (stop time, fixed step, maximum step) now read `param.*`, and the 9 values typed in the blocks (load, resistor of the neutral point, measuring dividers, data of the switches and diodes) moved to `param`. Table 3 lists the defects and their fixes.



Table 3: Defects of the model as saved and their fixes (November 2026).

part	as saved	now
DDSRF-PLL	stage A reads outputs 3 and 4 of `ddsrf` (negative sequence) through a gain −1; frequency integrator from 0, no clamp: the loop never locks on the grid phase	outputs 1 and 2 (positive sequence), gain +1; new Saturation 0.5 … 1.5 pu on the frequency integrator, which starts at 1 pu; gains 0.32 and 8 (Section 4.5)
modulator	the three lines into the Bus Creator of `modulator1` have no names, so the Bus Selectors of the gate commands find nothing	lines named `du(t)`, `dv(t)`, `dw(t)`, the names that the PWM block of the library selects
current PI	gains of the PMSM inverter of the template (`psm_ctrl.kp_i`, `ki_i`) and its sample time `ts_inv`	`param.afe.ctrl.kp_is_xi`, `ki_is_xi` (ξ axis), `kp_is_eta`, `ki_is_eta` (η axis) and t_s; same numbers
From tags	`gamma_grid_emulator`, `udc_mod1`, `phase_grid_dqpll`, `omega_grid_pu_dqpll`, `pwr_lim_enable`: no Goto	`phase_grid_emulator`, `udc`, `phase_grid_pll`, `omega_grid_pll_pu`; `pwr_lim_enable` is a Constant 0
template blocks	18 From blocks of a template bridge (thermal data, DC-link current, other PLLs, power meter of the template control) with no source	commented out, with the To Workspace blocks they alone feed
power meter	bus object `PowerMeterIecBus` not defined	created by the initialization
tag visibility	the Goto `Vgrid_unorm_ddsrf` of the DDSRF-PLL is scoped, and no Goto Tag Visibility shows it to its From in `pll_stageB`	Goto Tag Visibility `Vgrid_unorm_ddsrf` in `afe_control` (added in MATLAB)

The effect of the PLL wiring is shown in Figure 5 of Section 5.2: with the PLL as saved (and every other fix applied) the switched replica never locks, the converter current reaches 1383 A and the DC link 1916 V, and the grid current has a THD of 10.4 %. Ten Goto blocks have no reader (`Cdc1_current`, `np_current`, `iz_*`, …): they are harmless and were left.

4 Design check

The numbers of this section are printed by the initialization (`prv.vienna_design`) and stored in `param.afe.design`. All the loops are discrete, with the period t_s, the one-step delay of the acquisitions and the zero-order hold of the modulator.

4.1 Operating point

With the load and the measuring dividers, the DC power is $P = u_{dc}^2/R + u_{dc}^2/(2R_{meas}) = 229.0\,\text{kW}$ at 1070 V. The control keeps the converter current in phase with the capacitor voltage ($i_\eta = 0$, which the PLL aligns to v_c), so with the phasors of the LCL filter at the secondary of the transformer

$$E\,e^{j\varphi} = V_c + Z_2\big(I_1 + Y_c V_c\big), \qquad V_{conv} = V_c - Z_1 I_1, \qquad P = \tfrac{3}{2}\,\mathrm{Re}\big(V_{conv} I_1^*\big), \tag{4.1}$$



with V_c and I_1 real, $Z_1 = R_1 + j\omega L_1$, $Z_2 = R_2 + j\omega L_2$ and $Y_c = 1/(R_c + 1/(j\omega C_f))$. Table 4 gives the solution. The converter voltage needs 0.913 of the linear range of the modulation, so 8.7 % of margin is left for the transients and for a grid voltage above nominal (the current PI clips at $|u| = 1$). It lags the converter current by $\delta = 4.34°$, the angle that sets the window of Section 4.6.

Table 4: Operating point at 229 kW (phase peaks).

quantity	value	note
no-load voltage E	569.2 V	at the secondary
capacitor voltage V_c	563.5 V	1.000 pu
converter current I_1	271.4 A	0.826 pu
grid current $\lvert I_2\rvert$	273.8 A	power factor 0.992 at the capacitor
converter voltage $\lvert V_{conv}\rvert$	564.0 V	0.913 pu of $u_{dc}/\sqrt{3}$, margin 8.7 %
angle δ	4.34°	converter voltage behind the current

4.2 LCL resonance

The resonance of the filter is

$$f_{res} = \frac{1}{2\pi}\sqrt{\frac{L_1 + L_2}{L_1 L_2 C_f}} = 1381\,\text{Hz}, \tag{4.2}$$

just above $f_s/6 = 1333$ Hz. A loop on the converter current with 1.5 periods of delay (one for the acquisition, half for the hold) adds damping to a resonance below $f_s/6$ and takes damping away above it [6, 7]. Here the passive damping of the mode, given by R_c, is 5.9 %, and with the current loop closed it is 5.8 % at 1390 Hz: the loop takes a little and R_c keeps the mode stable. The grid side is only the leakage of the transformer; a stiffer grid (a larger transformer) moves the resonance further above $f_s/6$, an additional grid inductance moves it below, where the loop damps it.

4.3 Current loop

The vector PI of the model integrates in the $\xi\eta$ frame, which turns at ω. In the stationary frame, with complex quantities $x = x_\alpha + jx_\beta$, the same controller is

$$C(z) = k_p + k_i\, t_s \frac{z}{z - \rho}, \qquad \rho = e^{j\omega t_s}, \tag{4.3}$$

an integrator with its pole at $+\omega$: infinite gain for the positive sequence at the grid frequency, and for the negative sequence at $-\omega$ only the finite gain $k_i/(2\omega) = 0.056$, a fifth of k_p. The plant is the LCL filter from the converter voltage to i_1, with the states $[i_1,\ v_c,\ i_2]$, discretized with the zero-order hold over t_s (A_d, B_d). With $g = (u_{dc}/\sqrt{3})/i_{bez}$, the gain from a current error in



amperes to the converter voltage in volts, and the sample one period old, the open loop is

$$L(z) = C(z)\,\frac{1}{z}\,g\,[1\ 0\ 0]\,(zI - A_\mathrm{d})^{-1}B_\mathrm{d}, \tag{4.4}$$

and the closed loop has the states $[i_1,\ v_\mathrm{c},\ i_2,\ i_1(k-1),\ X(k-1)]$, with X the integrator: $u(k) = (k_\mathrm{p} + k_\mathrm{i}t_\mathrm{s})(r - i_1(k-1)) + \rho X(k-1)$, $X(k) = \rho X(k-1) + k_\mathrm{i}t_\mathrm{s}(r - i_1(k-1))$. The positive sequence is read on $L(e^{j2\pi f t_\mathrm{s}})$ for $f > 0$, the negative on $f < 0$.

Table 5 and Figure 3 give the result: crossover at 135 Hz with 70° of phase margin for the positive sequence and 131 Hz with 78° for the negative sequence, gain margin 16.3 dB (the phase crosses −180° just before the resonance). The slowest closed-loop pole in the $\xi\eta$ frame is $-126 + 64j$ rad/s (the loop has complex coefficients there, so its poles do not come in conjugate pairs) (7.9 ms). The loop becomes unstable at $k_\mathrm{p} = 1.67$: the gain is limited by the delay, and $k_\mathrm{p} = 0.25$ leaves a factor of 6.7.

Table 5: Linear checks of the loops (`vienna_design`).

loop	gains	crossover	PM	GM	note
current, positive sequence	k_p 0.25, k_i 35	134.7 Hz	69.6°	16.3 dB	unstable from $k_\mathrm{p} = 1.67$
current, negative sequence	same	131.2 Hz	77.6°	16.3 dB	finite gain at $-\omega$
LCL mode with the loop		1390 Hz			damping 5.8 % (passive 5.9 %)
DC voltage	k_p 4, k_i 45	21.1 Hz	97.5°	29.0 dB	load pole 55.6 rad/s
DDSRF-PLL with its filter	k_p 0.32, k_i 8	15.7 Hz	58.3°		slowest pole −37 rad/s

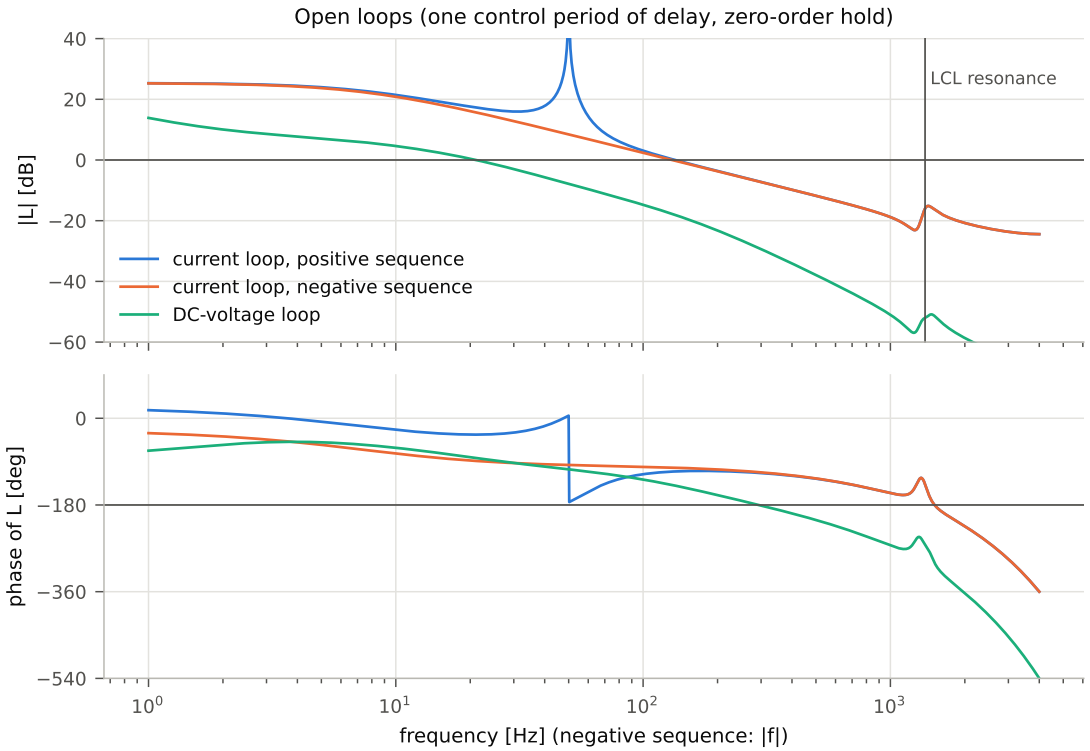


Figure 3: Open loops: current loop (4.4) for the positive and the negative sequence (the positive one has the integrator pole at 50 Hz) and DC-voltage loop (4.6). The vertical line marks the LCL resonance.

4.4 DC-voltage loop

The DC link seen by u_{dc} is $C_{\mathrm{ser}} = 7.2\,\mathrm{mF}$ with the resistive load. Linearized at the operating point, one pu of active current changes u_{dc} (in pu) as

$$G_{\mathrm{u}}(s) = \frac{K}{s+a}, \qquad K = \frac{\frac{3}{2}\,\mathrm{Re}(V_{\mathrm{conv}})\,i_{\mathrm{bez}}}{C_{\mathrm{ser}}\,u_{\mathrm{dc0}}\,u_{\mathrm{dc,nom}}} = 33.6/\mathrm{s}, \qquad a = \frac{2}{C_{\mathrm{ser}}}\left(\frac{1}{R} + \frac{1}{2R_{\mathrm{meas}}}\right) = 55.6\,\mathrm{rad/s}, \tag{4.5}$$

where the factor 2 of the pole comes from the load power u_{dc}^2/R. The loop is

$$L_{\mathrm{u}}(z) = \left(k_{\mathrm{p}} + k_{\mathrm{i}}\,t_{\mathrm{s}}\frac{z}{z-1}\right) T_{\mathrm{dd}}(z)\,G_{\mathrm{u}}(z)\,\frac{1}{z}, \tag{4.6}$$

with $G_{\mathrm{u}}(z)$ the zero-order-hold equivalent of $G_{\mathrm{u}}(s)$ and T_{dd} the closed current loop from the ξ reference to the ξ current, obtained from the stationary-frame closed loop T as $T_{\mathrm{dd}}(f) = \big(T(f+f_0) + \overline{T(-f+f_0)}\big)/2$. The loop crosses at 21.1 Hz with 97.5° of phase margin: a well-damped and slow loop, about six times slower than the current loop. Its speed matters for the start-up (Section 5.2) and for load steps: the step from no load to the full load takes the DC link down to about 910 V in the switched replica (Table 7). Below about 980 V the converter voltage of the full load, 564 V phase peak, is beyond the linear range of the modulation, $u_{\mathrm{dc}}/\sqrt{3}$ (at 910 V it would need $|u| = 1.07$): the current PI clips, the bridge rectifies like a diode bridge, and u_{dc} recovers from such a dip in 0.25 s to 0.4 s, much slower than the loop.



4.5 DDSRF-PLL

With the positive-sequence feed of Table 3, the error of the PLL is the η component of the decoupled positive sequence after its low-pass filter $F(s) = 1/(\tau s + 1)$, $\tau = 1/\omega$, normalized by the voltage. For small errors the loop is

$$L_{\mathrm{pll}}(s) = \frac{\omega(k_{\mathrm{p}} s + k_{\mathrm{i}})}{s^2(\tau s + 1)}, \qquad \tau s^3 + s^2 + \omega k_{\mathrm{p}}\, s + \omega k_{\mathrm{i}} = 0, \tag{4.7}$$

the second order loop of the PLL with the filter in it [3]. The gains of `ctrl_afe_setup`, $k_{\mathrm{p}} = 1$, $k_{\mathrm{i}} = 80$, give a damping of 0.99 without the filter but only 33.5° of phase margin with it. Table 6 compares three sets; the model uses 0.32 and 8, the set of `frt_dqpll` after its fix, with the best margin. The lock times of the switched replica (error below 5° from then on) differ less than the margins because the first 12.5 ms are spent with the trackers in reset; the start-up current is about the same for the three sets.

Table 6: Gain sets of the DDSRF-PLL: natural frequency and damping without the filter, crossover and phase margin with the filter, lock time in the switched replica.

gain set	k_{p}	k_{i}	ω_{n} [rad/s]	ζ	crossover, PM	lock
`ctrl_afe_setup`	1	80	158.5	0.99	40.7 Hz, 33.5°	62 ms
model (as `frt_dqpll`)	0.32	8	50.1	1.00	15.7 Hz, 58.3°	44 ms
faster	0.6	30	97.1	0.97	27.4 Hz, 45.1°	38 ms

The initial value of the frequency integrator matters more than the gains. From zero, as in `frt_dqpll`, the clamp holds the integrator at 0.5 pu, the PLL locks only after 112 ms, and in the meantime the current loop works in a frame that turns at half the grid frequency: the start-up current reaches 1228 A and the DC link 1513 V (Figure 5). Started at 1 pu, the PLL locks at 44 ms.

4.6 Vienna constraint and neutral point

Window of the zero crossings. Over one period of the operating point, the pole reference of a phase and its current have opposite signs where the converter voltage, which lags the current by δ, has not yet crossed zero: about δ after each zero crossing of the current. With the min-max zero sequence (2.4) this window is 8.60° per period, close to $2\delta = 8.68°$ (Figure 4). In the window the leg stays at the neutral point (2.3) and the converter voltage of that phase misses a few volts: this distorts the current near its zero crossings (Section 5.3).

The zero sequence is free in a three-wire system and can be used to give each pole reference the sign of its current: within the limits $|m| \le 1$, choose the zero sequence v_0 closest to the min-max one that satisfies $\mathrm{sign}(m_{\mathrm{x}}) = \mathrm{sign}(i_{\mathrm{x}})$ for the three phases. At the operating point such a zero sequence exists over 98 % of the period and the window falls to 2.20° per period. This is not in the model: it needs a change of `sv_pwm_cm` (C code of the library) or a block after it; the switched replica tests it in Section 5.3.

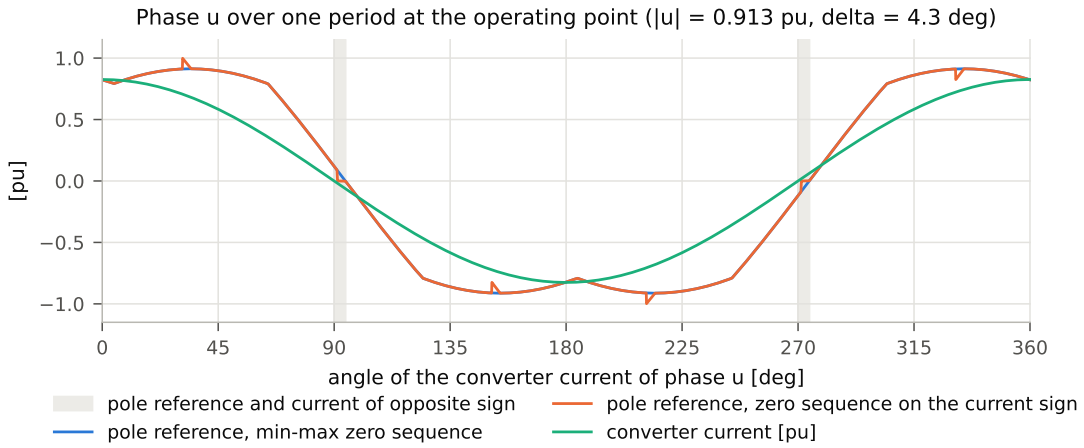


Figure 4: Phase u over one period of the operating point: pole reference with the min-max zero sequence, pole reference with the zero sequence that follows the current sign, converter current. The two references differ only near the zero crossings of the three currents; the shaded windows are where the min-max reference and the current have opposite signs.

Neutral point. A leg is at the neutral point for the fraction $1 - |m_{\mathrm{eff}}|$ of each carrier period, so the current into the neutral point and the voltage difference of the two halves are

$$i_{\mathrm{np}} = \sum_{x=u,v,w} \left(1 - |m_{\mathrm{eff,x}}|\right) i_{\mathrm{x}}, \qquad \frac{d}{dt}\left(v_{\mathrm{C2}} - v_{\mathrm{C1}}\right) = \frac{2\, i_{\mathrm{np}}}{C_{\mathrm{np}}} \tag{4.8}$$

(the measuring dividers are neglected). At the operating point i_{np} has no mean value and a third harmonic of 33.0 A (peak 43.2 A), which gives a ripple of $v_{\mathrm{C2}} - v_{\mathrm{C1}}$ of 5.4 V peak to peak at 150 Hz, 0.5 % of u_{dc}. The switched replica gives the same ripple and a mean of +0.3 V (Figure 10): there is no drift, and no balancing loop is needed in this symmetric case.

5 Switched replica: start-up and steady state

5.1 The replica

`py/vienna_sim.py` (numpy and numba) integrates the power stage with the step $t_{\mathrm{c}} = 0.625\,\mu\mathrm{s}$ of the model (semi-implicit Euler) and runs the control at every peak and valley of the carrier, as the model does:

- grid emulator with its ramp and second order filter; transformer reduced to its leakage and copper resistance at the secondary, with the no-load voltage E shifted by 30° (no magnetizing branch); LCL filter with the star of the capacitors floating; DC link floating, so the three converter currents add to zero;
- Vienna legs by (2.1): ideal switches and diodes; a phase whose path is blocked for the sign it would take stays at zero current (open phase) while its voltage lies between the two candidate pole voltages; gates by (2.2) with the turn-on delay of the dead time;
- control as Section 2.4: 12-bit acquisitions one step late, the first harmonic trackers of



the library C code, the DDSRF network and its loop, the DC-voltage PI with back-calculation, the vector PI with clipping, the inverse Park transform and `sv_pwm_cm` with the sign inversion of the modulator;

- options for the comparison, not in the model: the DDSRF-PLL as saved, the initial value of the PLL integrator, no dead time, the zero sequence on the current sign, the lower limit and the anti-windup of the DC-voltage PI, the initial value of the DC-voltage reference, a precharge voltage, a load step, a feed-forward of the capacitor voltage.

Not modelled: the clock alignment (one clock), the power meter and the logging, the conduction drops of the devices (diode 2 V, switch 1.5 mΩ) and their RC snubbers. Table 7 gives the cases and Table 8 the start-up options; each case runs 1.25 s.

Table 7: Switched replica: start-up and steady state. $\hat{\imath}$: largest converter current and when; u_{dc} max and min (min over the first 0.3 s); 2 %: last time u_{dc} is out of 1070 V ± 2 %; lock: PLL error below 5° from then on; THD of the grid and converter currents over the last two periods; Δv_{np}: peak-to-peak ripple of $v_{C2} - v_{C1}$; window: degrees per period where the pole reference and the current have opposite signs.

case	$\hat{\imath}$		u_{dc} [V]		2 %	lock	THD [%]		Δv_{np}	window
	[A]	[ms]	max	min	[ms]	[ms]	grid	conv.	[V]	[deg]
as saved: PLL on the negative sequence	1383	119	1916	784	1250	no	10.44	9.92	54.5	60.6
fixed PLL, integrator from 0	1228	28	1513	760	229	112	3.42	2.80	5.4	11.0
model as fixed	**752**	**20**	**1171**	**758**	**288**	**44**	**3.41**	**2.79**	**5.4**	**11.0**
no dead time	752	20	1170	760	288	44	3.00	2.34	5.6	10.5
zero sequence on the current sign	751	20	1170	758	289	44	2.47	1.73	6.0	2.5
load connected at 0.2 s	318	219	1341	913	577	50	3.42	2.81	5.4	11.0
the same, feed-forward, reference from 1 pu	318	213	1125	906	443	50	3.44	2.93	5.4	9.5

5.2 Start-up

Figure 5 compares the model as saved (PLL on the negative sequence), the fixed PLL started from zero and the model as fixed. As saved, the phase error of the PLL never stops turning, the current loop works in a wrong frame and the DC link oscillates between 1.6 and 1.9 kV. With the fixed PLL started at 1 pu the model starts in about 0.3 s, but with a current peak of 752 A (2.3 pu) at 20 ms. Figure 6 shows how it builds up:

- for the first 15 ms the grid voltage is still small and the load drains the DC link with $RC_{ser} = 36$ ms: u_{dc} falls to 758 V, below the level of the diodes, $\sqrt{3}E = 986$ V;
- the DC-voltage reference rises from zero (step at 1 ms and 20 ms filter) and stays below u_{dc} for the whole start, so the error e of the DC-voltage PI is negative. The PI is in velocity form with back-calculation: it stores $x = i_\xi^* - k_p e$ also while its output is clamped, so each step adds $k_p \Delta e + k_i t_s e$ to the last, clamped output. The error starts at −1 pu, the output at its lower limit, −1.65 pu, and the stored state is $-1.65 + 4 = +2.35$ pu. As the reference rises and u_{dc} falls, the error shrinks and the output rises with it: it turns positive at 8 ms and reaches 1.16 pu at 15 ms, with u_{dc} still 220 V above its reference. The PI asks for an active current that the DC voltage does not need;
- the current PI has no feed-forward of the grid voltage: its integrators must build the whole

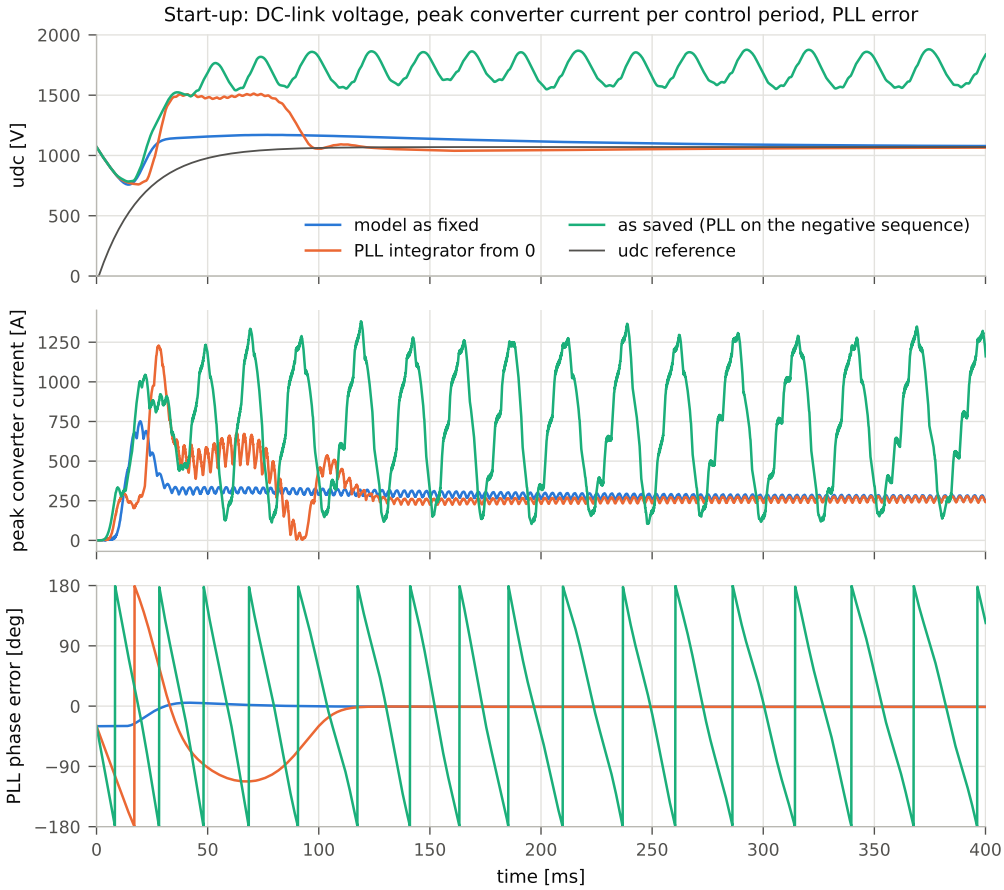


Figure 5: Start-up in the switched replica: DC-link voltage with the reference of the model, largest converter current per control period, phase error of the PLL. Model as fixed, fixed PLL with the integrator started from 0, PLL as saved.

converter voltage while the grid voltage rises, and the current overshoots its reference, 2.26 pu (743 A) at 20 ms against 1.14 pu. This is the peak;

- the current then recharges the DC link above its reference, to 1171 V (+9.4 %), and the bridge cannot give the energy back: only the load brings u_{dc} down, within 2 % after 288 ms.

Item 7 of the design check prints the threshold of the second point: the output of the PI turns positive when the error rises above $-x/(k_p + k_i t_s) = -0.59$ pu (-0.52 pu in the replica, where the integral of the first steps adds to the threshold).

Table 8 and Figure 7 give the options. Each of the two working changes removes one of the two causes; each half of a change alone is not enough.

- **DC-voltage PI with lower limit 0 and conditional integration.** The bridge cannot give a negative active current, so 0 is the right lower limit; with conditional integration the integrator holds while the output is clamped, in place of the back-calculation. The reference of the current stays at zero until 41 ms, when u_{dc} has fallen below the rising reference (Figure 6). The peak falls to 456 A and u_{dc} does not overshoot. These 456 A are the current that the rising grid voltage drives through the filter while the current loop, still without feed-forward, holds its zero reference only with a large error. With the lower

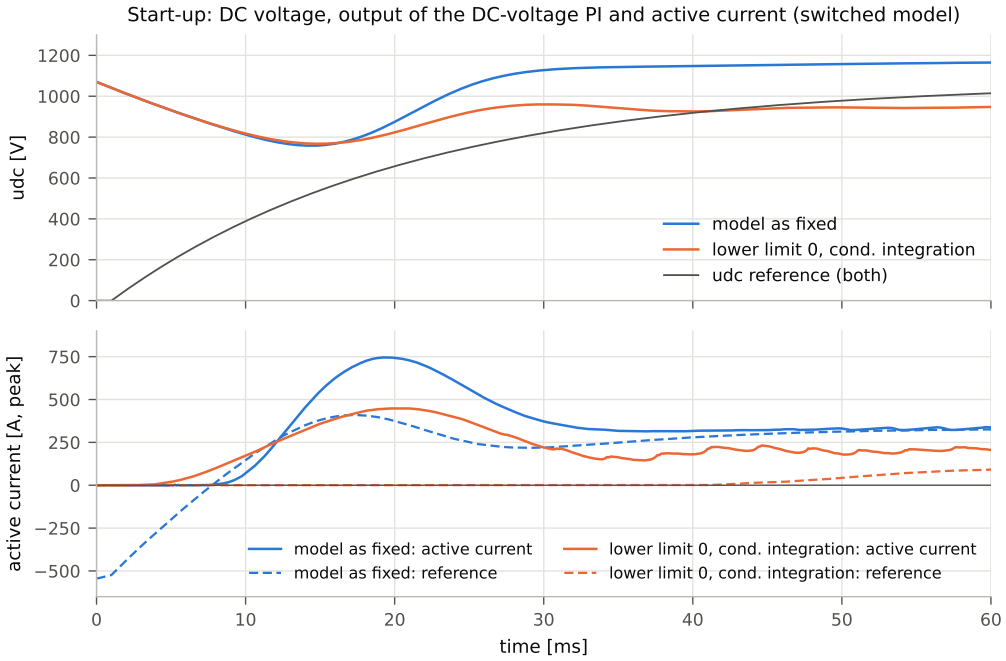


Figure 6: First 60 ms of the start-up in the switched replica, model as fixed and DC-voltage PI with lower limit 0 and conditional integration: DC-link voltage and its reference (top); active current and its reference, the output of the DC-voltage PI (bottom).

limit at 0 and the back-calculation kept, the output starts its rise from 0 instead of −1.65 pu and the peak is 1068 A; with conditional integration and the limit kept at −1.65 pu, the reference of the current stays negative until 50 ms, which the bridge cannot follow: the DC link falls to 647 V and the diodes recharge it at full grid voltage (587 A).

- **Feed-forward of the capacitor voltage and DC-voltage reference from the measured voltage.** The feed-forward $u_{\mathrm{ff}} = -(\sqrt{3}\,u_{\mathrm{bez}}/u_{\mathrm{dc}})\,v_{\mathrm{c}}$ on the output of the current PI, in the $\xi\eta$ frame with the sample advanced by $1.5\,t_{\mathrm{s}}$, lets the current follow its reference. With the reference of u_{dc} started from the measured voltage (here 1 pu), the DC-voltage PI asks for current only to recharge the DC link, up to its limit of 1.65 pu (542 A): the peak is 455 A and u_{dc} does not overshoot. The steady state is the same (THD 3.43 %): with this filter the feed-forward does not upset the damping of the LCL mode. The reference from 1 pu without the feed-forward gives 782 A, the feed-forward with the reference from zero 511 A and an overshoot to 1185 V.

The two changes together give the same as the second one alone. The feed-forward with the reference from zero should not be combined with the changed PI: the current then stays near zero during the ramp of the grid voltage, the load drains the DC link to 611 V, and the recharge at full grid voltage reaches 677 A.

With either change u_{dc} comes back to its reference slowly, within 2 % after 317 ms and 235 ms: below about 980 V the current PI clips (Section 4.4) and the bridge rectifies like a diode bridge. The dip to about 770 V is the load draining the DC link while the grid voltage rises from zero, a condition of the test rather than of the converter; a precharge to 980 V changes little. Connecting

the load later does not help either (Table 7): without the feed-forward the start-up current alone charges the unloaded DC link to 1341 V; with it and the reference from 1 pu the start draws less than 30 A, but the DC link still creeps up to 1125 V in 0.2 s. Near zero load the current cannot be held at zero: the short current pulses that remain are rectified into the DC link (about 1.8 kW here) and nothing takes the energy back, so a Vienna rectifier needs a minimum load to hold its DC voltage.

Table 8: Start-up options in the switched replica, each a change of the model as fixed. Columns as in Table 7; u_{dc} max 1070 V is the precharge: no overshoot.

change	$\hat{i}$ [A]	[ms]	u_{dc} [V] max	min	2 % [ms]	THD grid [%]
none (model as fixed)	752	20	1171	758	288	3.41
DC-voltage PI: lower limit 0, conditional integration	**456**	**19**	**1070**	**767**	**317**	**3.39**
lower limit 0 only (back-calculation kept)	1068	17	1251	763	338	3.41
conditional integration only (limit −1.65 pu kept)	587	26	1070	647	346	3.42
DC-voltage reference from 1 pu	782	16	1084	785	263	3.42
feed-forward, reference from 0	511	23	1185	704	307	3.42
feed-forward, reference from 1 pu	**455**	**17**	**1070**	**772**	**235**	**3.43**
feed-forward and the changed PI, reference from 0	677	27	1070	611	306	3.44
DC link precharged to 980 V	724	20	1136	701	249	3.40

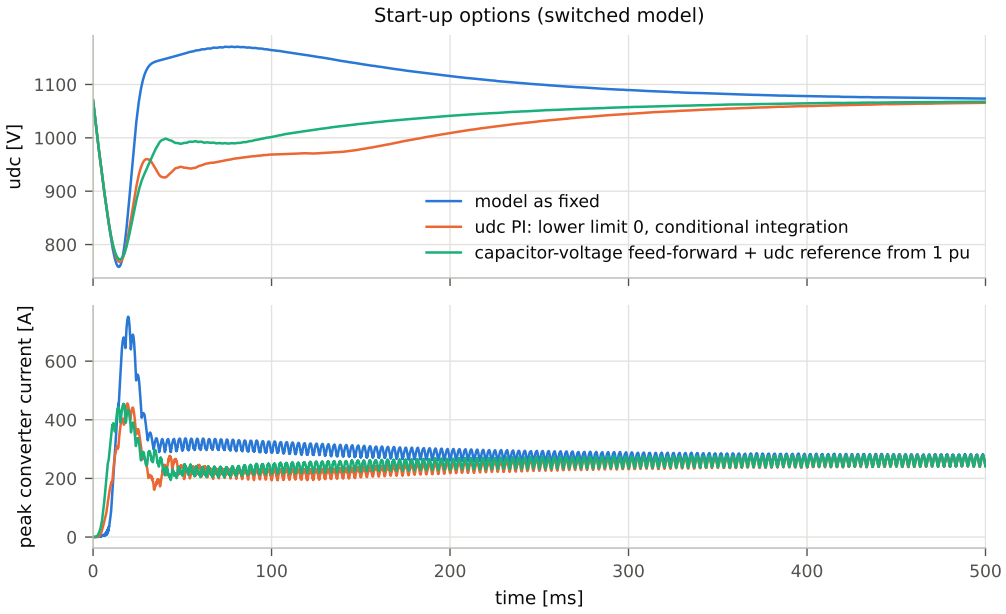


Figure 7: Start-up options in the switched replica: model as fixed, DC-voltage PI with lower limit 0 and conditional integration, feed-forward of the capacitor voltage with the DC-voltage reference started from the measured voltage.

5.3 Steady state

In steady state (Figure 8) the grid current has a fundamental of 275.1 A peak and a THD of 3.41 %, with 2.0 % of 5th, 1.6 % of 7th, 1.1 % of 11th and 0.9 % of 13th harmonic (Figure 9); the reactive current is zero. The low-order harmonics come mostly from the zero crossings: the

converter current minus its fundamental shows a step at each zero crossing of the current, where the leg stays at the neutral point and a phase is open for a short time (one phase open 1.24 % of the time). The window of the replica is 11.0° per period, a little more than the 8.6° of the linear estimate, because the switching ripple makes the current change sign several times near its zero crossings. The dead time adds 0.4 % of THD.

The zero sequence on the current sign (Section 4.6), computed with the sign of the reference current $1.5\,t_s$ ahead, reduces the window to 2.5° per period and the THD to 2.47 % (5th 1.2 %, 7th 0.9 %), with the same start-up and a neutral-point ripple of 6.0 V instead of 5.4 V.

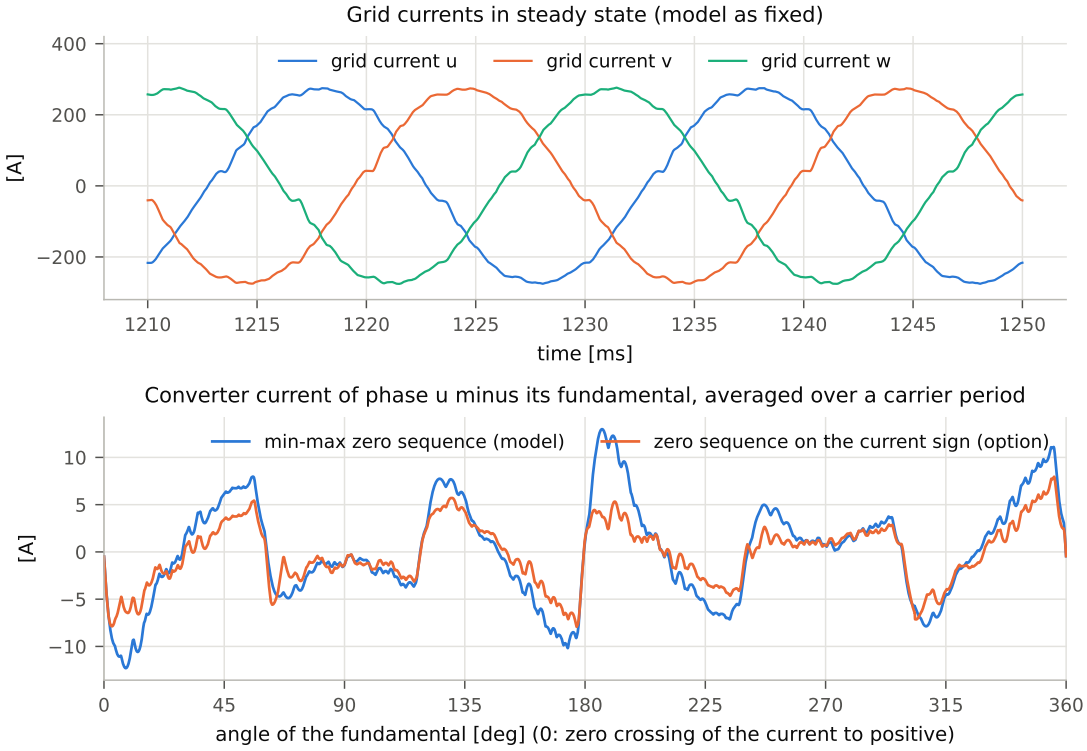


Figure 8: Steady state in the switched replica. Top: grid currents. Bottom: converter current of phase u minus its fundamental, averaged over a carrier period, over one period from its positive-going zero crossing, with the min-max zero sequence of the model and with the zero sequence on the current sign.

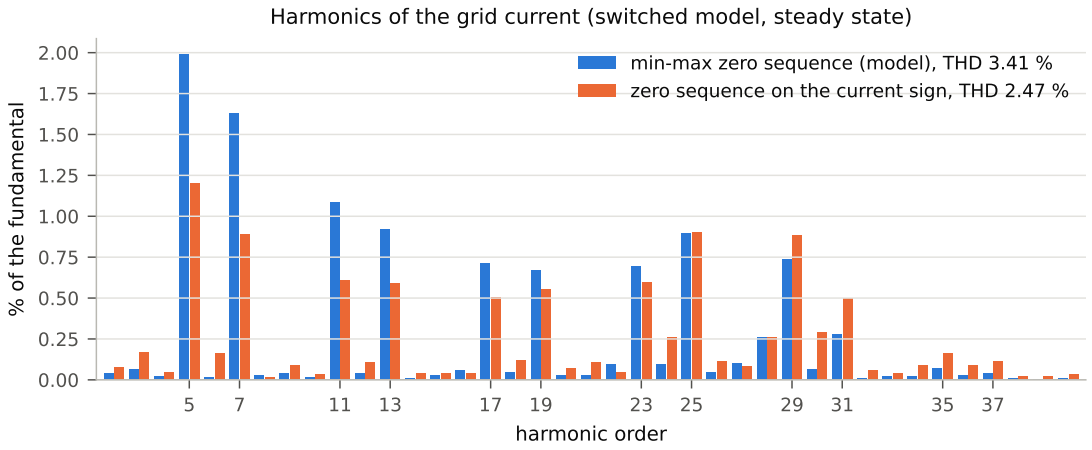


Figure 9: Harmonics of the grid current in steady state (switched replica, two periods), in % of the fundamental.

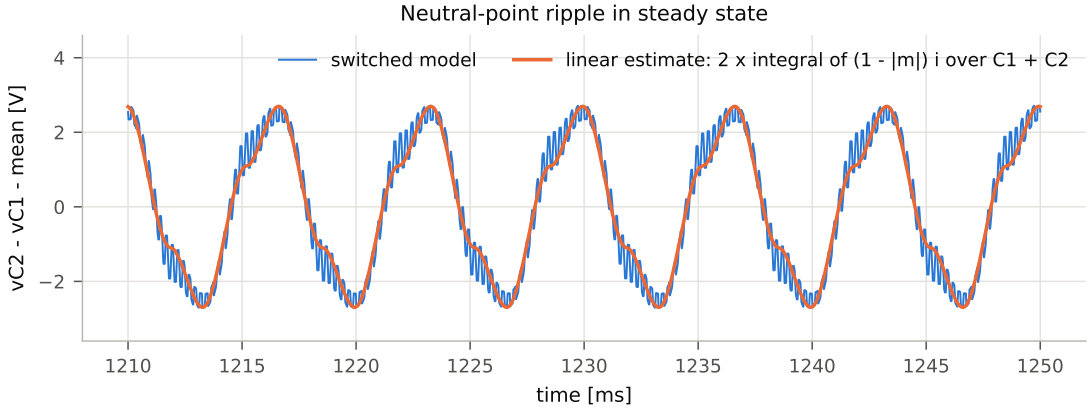


Figure 10: Ripple of $v_{C2} - v_{C1}$ in steady state: switched replica and linear estimate (4.8).

6 Open points and recommendations

1. **Run the revised model in MATLAB.** The model was opened in MATLAB and its scoped tag fixed (Table 3), but no run has been compared with the replica yet. The initialization prints the design check and the result of the two model checks; the start-up and the steady state of the model should match Table 7 (model as fixed).

2. **Start-up current** (Section 5.2). Recommended, one of two changes, each about 455 A instead of 752 A in the replica and no overshoot:

   (a) in `udc_ctrl`, the lower limit of the Saturation at 0 and conditional integration: the Unit Delay of the PI keeps its state while the output is clamped, in place of $i_\xi^* - k_\mathrm{p} e$. A change of one subsystem of the model, with the reference of the model unchanged;

   (b) a feed-forward of the measured capacitor voltage on the output of the current PI (sample advanced by $1.5\,t_\mathrm{s}$, scaled by $\sqrt{3}\,u_\mathrm{bez}/u_\mathrm{dc}$) and a DC-voltage reference that starts from the measured u_dc (the low-pass block of the library has no initial state,



so the filter must be initialized or replaced). The feed-forward also helps the current loop in grid transients.

Both together give the same as (b); do not combine (a) with the feed-forward while the reference still starts from zero (Table 8). Enabling the PWM only after the lock of the PLL is a further option. The rectifier also needs a minimum load: with no load the DC link creeps up.

3. **Voltage margin** (Section 4.4). At 1070 V the converter voltage of the full load needs 0.913 of the linear range of the modulation; below about 980 V it is out of range. After the start and after a step to the full load (dip to about 910 V) the current PI clips, the bridge rectifies like a diode bridge and u_{dc} recovers in 0.25 s to 0.4 s. A faster recovery needs a higher DC-voltage reference, if the devices allow it, or smaller load steps.

4. **Distortion at the zero crossings** (Sections 4.6 and 5.3). A zero sequence that follows the sign of the current lowers the THD of the grid current from 3.4 % to 2.5 %. It needs a change of `sv_pwm_cm` (C code of the library, used by other models) or a block between it and the modulator.

5. **LCL resonance** (Section 4.2). At 1381 Hz, just above $f_{\mathrm{s}}/6$, the mode keeps its passive damping (5.8 %). With a stiffer grid it moves up and the converter-current loop cannot damp it; if the filter or the transformer changes, check it again, or add active damping (capacitor current) or move the loop to the grid current.

6. **Neutral point.** There is no balancing loop: the zero-sequence PI is made for modules in parallel and its output gain is 0. In this symmetric case the mean unbalance is 0.3 V; with unbalanced grids, unequal capacitors or unequal devices a loop on the zero sequence will be needed. The zero sequence on the current sign uses the same degree of freedom: the two must be designed together.

7. **Fault ride-through.** The data of test 25.4 of VDE-AR-N 4110 are computed but not applied, and the Vienna control has no FRT logic; the DDSRF-PLL is ready for asymmetric dips [3], the rest of the control is not.

8. **Housekeeping.** Ten Goto blocks have no reader; `init_model_legacy.m` can be deleted once the new initialization is adopted.

7 Files and how to reproduce

- The model folder `vienna_rectifier`, in `foundations_of_power_electronics` of [1]:
  - `vienna_rectifier.slx`, the model;
  - `init_model.m`, the initialization: run it from this folder, then run the model;
  - `vienna_rectifier_private.m` and the functions of `private/`: the parameter functions, the design check `vienna_design`, the bus of the power meter, the path of the library and the two model checks;



  - `init_model_legacy.m`, the old initialization, and the README with the list of the legacy names.

- This folder:
  - `py/vienna_sim.py`: the switched replica, `run(...)` with the options of Section 5.1;
  - `py/vienna_design_check.py`: the linear checks of `vienna_design` and the cases of Tables 7 and 8. Run `python3 vienna_design_check.py ../figures` from `py/` (a few seconds with numba): it writes every figure of this note and the numbers to `figures/vienna_design_check_output.txt`.

- The library [2]: the Vienna bridge `vienna_rectifier_ideal_switch_based_model`, the three-level PWM, the C callers `sv_pwm_cm` and `first_harmonic_tracker`, the classes `ctrl_afe_setup`, `hw_afe_three_phase_setup` and `grid_three_phase_emulator`.

References

[1] pwr-control, *modelization-and-control*, repository, folder `foundations_of_power_electronics/vienna_rectifier` (model `vienna_rectifier.slx`, README with the design check of November 2026). https://github.com/pwr-control/modelization-and-control

[2] pwr-control, *library*, repository: semiconductor bridges, modulators, C callers (`sv_pwm_cm`, `first_harmonic_tracker`), setup classes. https://github.com/pwr-control/library

[3] D. Bagnara, *Three-phase grid PLLs under grid faults: dq-PLL, DDSRF-PLL and FHT-PLL of the model frt_dqpll*, pwr-control technical note, rev. 00, November 2026, `repos_doc/three_phase_plls_comparison` of *repositories-documentation*.

[4] J. W. Kolar and F. C. Zach, "A novel three-phase utility interface minimizing line current harmonics of high-power telecommunications rectifier modules," *IEEE Transactions on Industrial Electronics*, vol. 44, no. 4, pp. 456–467, 1997.

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